Input system:
Dependent variables:x_1, x_3, x_2
Independent variables:t_1, t_2

$$\frac{\partial x_1}{\partial t_1} = (a_1 x_1^2 + b_1 x_2 + c_1)\sin(\mathrm{Hb}(x_1, x_2, x_3)) + (c_1 x_3^2 + b_1 x_1 + a_1)\cos(\mathrm{Hb}(x_1, x_2, x_3))$$

$$\frac{\partial x_1}{\partial t_2} = (a_2 x_1^3 + b_2 x_3 + c_2)\sin(\mathrm{Hb}(x_1, x_2, x_3)) + (c_2 x_3^2 + b_2 x_2 + a_2)\cos(\mathrm{Hb}(x_1, x_2, x_3))$$

$$\frac{\partial x_2}{\partial t_1} = (a_3 x_1^2 + b_3 x_2 + c_3)\sin(\mathrm{Hb}(x_1, x_2, x_3)) + (c_3 x_3^2 + b_3 x_1 + a_3)\cos(\mathrm{Hb}(x_1, x_2, x_3))$$

$$\frac{\partial x_2}{\partial t_2} = (a_4 x_1^3 + b_4 x_3 + c_4)\sin(\mathrm{Hb}(x_1, x_2, x_3)) + (c_4 x_3^2 + b_4 x_2 + a_4)\cos(\mathrm{Hb}(x_1, x_2, x_3))$$

$$\frac{\partial x_3}{\partial t_1} = (a_5 x_1^2 + b_5 x_2 + c_5)\sin(\mathrm{Hb}(x_1, x_2, x_3)) + (c_5 x_3^2 + b_5 x_1 + a_5)\cos(\mathrm{Hb}(x_1, x_2, x_3))$$

$$\frac{\partial x_3}{\partial t_2} = (a_6 x_1^3 + b_6 x_3 + c_6)\sin(\mathrm{Hb}(x_1, x_2, x_3)) + (c_6 x_3^2 + b_6 x_2 + a_6)\cos(\mathrm{Hb}(x_1, x_2, x_3))$$

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Result:
Dependent variables:$s_3, s_4, x_3, x_1, s_2, s_1, x_2$
Independent variables:t_1, t_2
Additional variables:

$$s_1 = \mathrm{Hb}(x_1, x_2, x_3)$$

$$s_2 = \mathrm{Hbi}(x_1, x_2, x_3)$$

$$s_3 = \sin(s_1)$$

$$s_4 = \cos(s_1)$$

$$\frac{\partial x_1}{\partial t_1} = s_3 a_1 x_1^2 + s_4 c_1 x_3^2 + s_3 b_1 x_2 + s_4 b_1 x_1 + s_3 c_1 + s_4 a_1$$

$$\frac{\partial x_1}{\partial t_2} = s_3 a_2 x_1^3 + s_4 c_2 x_3^2 + s_3 b_2 x_3 + s_4 b_2 x_2 + s_3 c_2 + s_4 a_2$$

$$\frac{\partial x_2}{\partial t_1} = s_3 a_3 x_1^2 + s_4 c_3 x_3^2 + s_3 b_3 x_2 + s_4 b_3 x_1 + s_3 c_3 + s_4 a_3$$

$$\frac{\partial x_2}{\partial t_2} = s_3 a_4 x_1^3 + s_4 c_4 x_3^2 + s_3 b_4 x_3 + s_4 b_4 x_2 + s_3 c_4 + s_4 a_4$$

$$\frac{\partial x_3}{\partial t_1} = s_3 a_5 x_1^2 + s_4 c_5 x_3^2 + s_3 b_5 x_2 + s_4 b_5 x_1 + s_3 c_5 + s_4 a_5$$

$$\frac{\partial x_3}{\partial t_2} = s_3 a_6 x_1^3 + s_4 c_6 x_3^2 + s_3 b_6 x_3 + s_4 b_6 x_2 + s_3 c_6 + s_4 a_6$$

$$\frac{\partial s_1}{\partial t_1} = (-1)s_1^3 s_2 s_3 a_3 x_1^2 + (-1)s_1^3 s_2 s_4 c_3 x_3^2 + (-1)s_1^3 s_2 s_3 b_3 x_2 + (-1)s_1^3 s_2 s_4 b_3 x_1 + (-1)s_1^2 s_2 s_3 a_5 x_1^2 + (-1)s_1^2 s_2 s_4 c_5 x_3^2$$

$$\frac{\partial s_1}{\partial t_2} = (-1)s_1^3 s_2 s_3 a_4 x_1^3 + (-1)s_1^3 s_2 s_4 c_4 x_3^2 + (-1)s_1^2 s_2 s_3 a_6 x_1^3 + (-1)s_1 s_2 s_3 a_2 x_1^3 + (-1)s_1^3 s_2 s_3 b_4 x_3 + (-1)s_1^3 s_2 s_4 b_4 x_2$$

$$\frac{\partial s_2}{\partial t_1} = 42s_3a_3x_1^2s_2^3s_1^8+42s_4c_3x_3^2s_2^3s_1^8+42s_3b_3x_2s_2^3s_1^8+42s_4b_3x_1s_2^3s_1^8+42s_3a_5x_1^2s_2^3s_1^7+42s_4c_5x_3^2s_2^3s_1^7+42s_3a_1x_1^2s_2^3s$$

$$\frac{\partial s_2}{\partial t_2} = 42s_3a_4x_1^3s_2^3s_1^8+42s_4c_4x_3^2s_2^3s_1^8+42s_3a_6x_1^3s_2^3s_1^7+42s_3a_2x_1^3s_2^3s_1^6+42s_3b_4x_3s_2^3s_1^8+42s_4b_4x_2s_2^3s_1^8+42s_4c_6x_3^2s_2^3s$$

$$\frac{\partial s_3}{\partial t_1} = (-1)s_4s_1^3s_2s_3a_3x_1^2+(-1)s_4^2s_1^3s_2c_3x_3^2+(-1)s_4s_1^3s_2s_3b_3x_2+(-1)s_4^2s_1^3s_2b_3x_1+(-1)s_4s_1^2s_2s_3a_5x_1^2+(-1)s_4^2s_1^2$$

$$\frac{\partial s_3}{\partial t_2} = (-1)s_4s_1^3s_2s_3a_4x_1^3+(-1)s_4^2s_1^3s_2c_4x_3^2+(-1)s_4s_1^2s_2s_3a_6x_1^3+(-1)s_4s_1s_2s_3a_2x_1^3+(-1)s_4s_1^3s_2s_3b_4x_3+(-1)s_4^2$$

$$\frac{\partial s_4}{\partial t_1} = s_3^2s_1^3s_2a_3x_1^2+s_3s_1^3s_2s_4c_3x_3^2+s_3^2s_1^3s_2b_3x_2+s_3s_1^3s_2s_4b_3x_1+s_3^2s_1^2s_2a_5x_1^2+s_3s_1^2s_2s_4c_5x_3^2+s_3^2s_1s_2a_1x_1^2+s_3s_1s_2s$$

$$\frac{\partial s_4}{\partial t_2} = s_3^2s_1^3s_2a_4x_1^3+s_3s_1^3s_2s_4c_4x_3^2+s_3^2s_1^2s_2a_6x_1^3+s_3^2s_1s_2a_2x_1^3+s_3^2s_1^3s_2b_4x_3+s_3s_1^3s_2s_4b_4x_2+s_3s_1^2s_2s_4c_6x_3^2+s_3s_1s_2s$$